

# Statement of Research

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## 1. Prior Research Experience: Efficient Structure Reasoning for Foundation Models

I'm Jianan Zhao, a fourth-year PhD candidate at Mila-UdeM. Before pursuing a PhD degree, I obtained a master degree at BUPT and was a visiting fellow at Microsoft Research Asia with a Star-of-Tomorrow award (Top-10%). During that period, I focused on graph structure learning, graph transformers, and knowledge-graph reasoning [8, 9, 10, 11, 12, 13], these works provided the graph-learning foundation for my PhD agenda. My PhD research focuses on a core question: how can foundation models reason over *structured* data *efficiently*? Scientific and biomedical data are rarely independent text snippets or tabular examples. They are graphs of entities, relations, hierarchies, and mechanisms. My work has therefore aimed to preserve structure while gaining the transferability, scalability, and flexibility of modern foundation models.

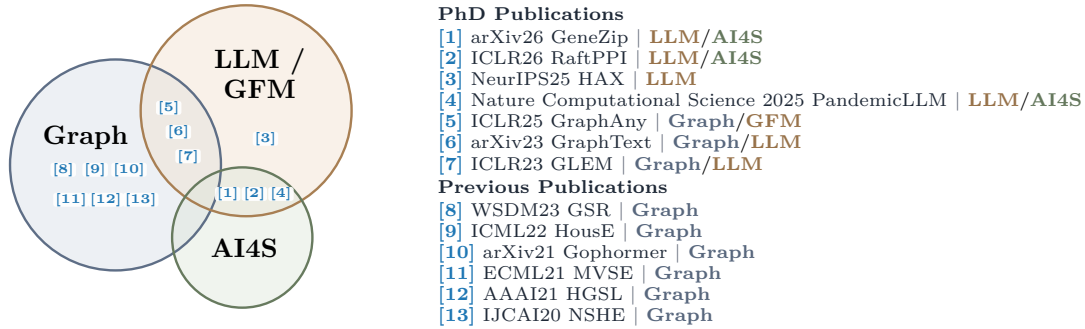


Figure 1: Selected publications across Graph, LLM / GFM, and AI4S (only first-, second-, or equal-contribution author works). Full list: [Google Scholar](#). Numbered markers and paper titles are clickable in the PDF.

Across these projects, I study efficiency across the foundation-model pipeline, spanning training, data, adaptation, and inference.

**Training efficiency: jointly using language and graph structure without expensive end-to-end training.** Large text-attributed graphs contain two complementary signals: text gives semantic meaning to nodes, while graph structure captures relations and community context. A direct cascade of a language model and a GNN is difficult to train jointly at scale because it couples two large learners with different optimization and memory profiles. In *GLEM* [7] (ICLR 2023 Notable, Top 5%), I developed a variational EM framework that lets a language model and a GNN improve one another through iterative co-training rather than costly end-to-end coupling. The result was both effective and efficient: it achieved first place on three Open Graph Benchmark node-property-prediction tasks at the time, while clarifying that graph-model/language-model interaction should be treated as a modeling problem, not merely a feature-concatenation trick.

**Data efficiency: training one model that transfers across many graphs.** Foundation models should not require a separate supervised model for each new graph. In *GraphAny* [5] (ICLR 2025), I worked on fully inductive node classification where test graphs can have unseen structures, feature spaces, and label spaces. *GraphAny* moved node classification closer to a true graph foundation model: trained on one graph, it surpassed GCN and GAT models trained separately on 31 target graphs, and completed training plus inference over those 31 graphs in roughly six minutes. This project strengthened my interest in reusable graph priors: instead of learning a new model for each graph, the goal is to learn transferable structural operations.

**Adaptation efficiency: using LLMs for graph tasks without graph-specific training.** My next step asked whether an LLM can perform graph reasoning if the graph is exposed through the right interface. In *GraphText* [6] (NeurIPS 2024 AFM Workshop), I represented local graph structure as a text-compatible graph syntax, enabling an LLM to operate on node-classification problems without additional graph-specific training. This was the first work, to my knowledge, showing that a training-free LLM can outperform standard supervised GNN baselines on node classification. The point was not that LLMs replace graph models; rather, *GraphText* showed that interface design can convert structured reasoning into a form where language models become useful collaborators.

**Inference efficiency: scaling foundation-model reasoning to scientific search spaces.** More recently, I have applied this efficiency perspective to AI for science. In *RaftPPI* [2] (ICLR 2026), we addressed proteome-scale protein-protein interaction retrieval. Brute-force screening is prohibitive at proteome scale, so we developed residue-level factorization that preserves prediction quality while reducing retrieval from 4.9 GPU months to 3.3 GPU minutes. In parallel, *GeneZip* [1] learned region-aware DNA compression, reaching 137.6× compression with

only +0.31 perplexity and a  $50.4\times$  eQTL speedup over JanusDNA. These projects moved my work from graph benchmarks toward scientific systems where structure, efficiency, and reliability matter simultaneously.

Together, these projects prepare me to contribute graph learning, foundation models, large-scale retrieval, and AI-for-science systems to Zitnik Lab, with a technical style centered on structural inductive bias, graph reasoning, and real biomedical deployment.

## 2. Future Research: From Edge Retrieval to Path Reasoning: PathProposer for LLMs over Biomedical Knowledge Graphs

My future research will focus on making LLMs better path reasoners over biomedical knowledge graphs [15, 16]. The key distinction is that the LLM should not be asked to *generate* or *search for* paths from scratch. Instead, a graph-native model should propose a compact set of top- $k$  concrete path instances, and the LLM should reason over these paths as explicit evidence. This distinction matters because biomedical questions are rarely answered by a single edge. A drug may act through a target, a pathway, a phenotype, and a disease mechanism; a gene-disease hypothesis may require connecting variants, proteins, tissues, and clinical concepts. Moving from one-hop edge retrieval to multi-hop path reasoning is therefore essential for mechanism-aware biomedical AI.

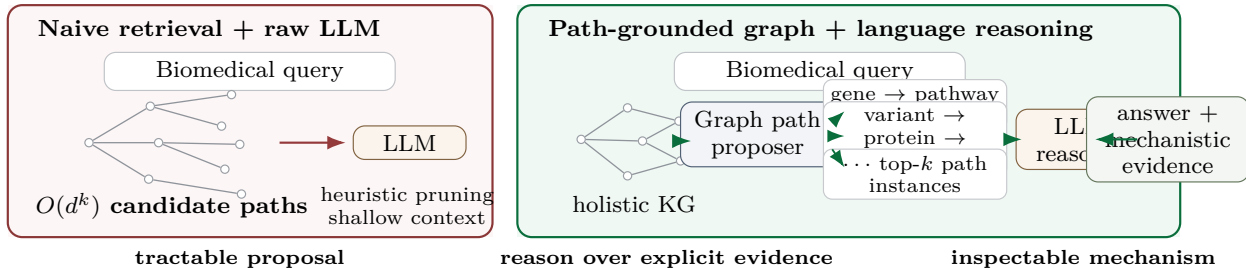


Figure 2: Path-grounded KG reasoning: a graph module proposes candidate paths, and the LLM reasons over explicit evidence instead of searching the KG directly.

A central starting point is *ARK* [17], which is highly relevant to this proposal and to Zitnik Lab. *ARK* shows that an LLM agent can balance breadth and depth by alternating between global lexical search over node descriptions and one-hop neighborhood exploration. This is an important step beyond shallow KG-RAG because one-hop operations can be composed into multi-hop exploration. However, it also exposes the next bottleneck. If the unit of retrieval remains an edge or a one-hop operation, path quality depends on inference-time search and on when the LLM prunes the frontier. Adding beam search can reduce the number of explored candidates from exhaustive  $O(d^L)$  enumeration to roughly  $O(LBd)$  under a fixed beam size  $B$ , but the path is still assembled through local tool decisions, often with expensive LLM calls and without a graph-native model trained on the full KG.

My proposal is to make the path, rather than the edge, the basic evidence object. *PathProposer* would take a query and a text-attributed biomedical KG, then return top- $k$  path instances such as drug-target-pathway-disease or variant-protein-phenotype-disease. The model would score paths through a factorized design: relation-level weights learn which biomedical relation sequences are plausible, while local transition scores select concrete next-hop entities and edges. Instead of materializing all length- $L$  paths, the proposer would run sparse dynamic programming or beam-style search over the KG, retaining backpointers so that the output is an explicit set of paths rather than only an answer node or a subgraph. With a fixed proposal budget, the computation becomes linear in hop length and local degree, while preserving multi-hop mechanisms for downstream reasoning.

This design changes the role of the LLM. The LLM is no longer the path generator or graph searcher; it is a path reasoner. Given a small set of proposed paths, the LLM can compare mechanisms, identify weak edges, integrate textual evidence, express uncertainty, and produce an answer grounded in inspectable graph evidence. Training can use multiple forms of supervision: known biomedical associations as weak endpoint labels, contrastive ranking between plausible and corrupted paths, *ARK*-style trajectories as distillation signals, and counterfactual tests that ask whether removing a key edge changes the final reasoning. These objectives would make the system not only accurate but also auditable.

I see this as a natural postdoctoral agenda in Zitnik Lab. *PathProposer* connects the lab’s interests in agentic AI, foundation models, geometric deep learning, large-scale knowledge graphs, LLMs, scientific discovery, and precision medicine: it is a graph-native component for biomedical reasoning that interfaces with LLMs while keeping mechanistic evidence explicit. I would evaluate it on biomedical KG question answering, target prioritization, therapy repurposing, and mechanism-aware hypothesis generation, while linking proposed paths to literature, omics, clinical records, and experimental tools. My long-term goal is knowledge-grounded foundation models that expose the paths supporting scientific claims, not just final answers.

### 3. References

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